

# Projection Relativity: Complete Master Equation Sheet

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## Purpose and Scope

This master equation sheet consolidates the current Projection Relativity (PR) framework into one detailed mathematical reference aligned with the main manuscript. Its purpose is to provide a complete reconstruction map for the theory: the variables, operators, projection sectors, recovery conditions, spectral constraints, geometric closure relations, and observational signatures are collected in one place.

The organizing principle is that observable physical structure emerges through constrained projections of a deeper internal spectral manifold. Gravity, Higgs structure, electromagnetism, strong-field continuum corrections, and cosmological response are treated as observable sectors of the same projection field.

$$\boxed{\Psi \rightarrow O_X \rightarrow \{u_n, \lambda_n\} \rightarrow P_i[\Psi] \rightarrow O_i} \quad (1)$$

This sheet uses the following status tags:

$D$  = derived from the framework,       $N$  = normalization or convention,       $C$  = constrained by recovery/stability,

## 1 Complete Variable and Parameter Dictionary

| Symbol          | Name                     | Meaning in PR                                                                                                                                                    | Tag |
|-----------------|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
| $\Psi(x, \xi)$  | Master projection field  | Underlying spectral field depending on observable spacetime coordinate $x$ and internal coordinate $\xi$ . All observable sectors are projections of this field. | D   |
| $x^\mu$         | Spacetime coordinate     | Effective low-energy coordinates in which projected observables are measured.                                                                                    | N   |
| $\xi, w$        | Internal coordinates     | Coordinates on the internal spectral manifold. In the reduced one-dimensional validation layer, $w$ is used.                                                     | D/N |
| $\mathcal{H}_P$ | Projection Hilbert space | Internal Hilbert space in which $\Psi$ , $O_X$ , eigenmodes, and projection operators are defined.                                                               | D   |

| Symbol                       | Name                               | Meaning in PR                                                                                                | Tag |
|------------------------------|------------------------------------|--------------------------------------------------------------------------------------------------------------|-----|
| $O_X$                        | Internal spectral operator         | Hamiltonian-like internal operator generating the spectral eigenbasis and gap structure.                     | D   |
| $V_X(w)$                     | Internal potential                 | Confining projection potential defining bounded spectral structure and finite-core recovery scale.           | C   |
| $u_n(w)$                     | Eigenmodes                         | Internal spectral modes satisfying $O_X u_n = \lambda_n u_n$ .                                               | D   |
| $\lambda_n$                  | Eigenvalues                        | Internal spectral values associated with $u_n$ .                                                             | D   |
| $\mu_n^2$                    | Excitation scales                  | Spectral gaps relative to the ground state; often $\mu_n^2 = \lambda_n - \lambda_0$ .                        | D   |
| $\mu_{\min}^2$               | First spectral gap                 | Minimum internal excitation scale $\lambda_1 - \lambda_0$ ; controls finite-core and propagator suppression. | D/C |
| $P_i[\Psi]$                  | Projection operators               | Sector-specific maps from the master field to observable structures.                                         | D   |
| $O_i$                        | Observables                        | Projected physical observables such as $g_{\mu\nu}$ , $H$ , $A_\mu$ , $\Sigma_X$ , and $H_{\text{eff}}$ .    | D   |
| $P_g$                        | Gravity projection                 | Stiffness projection producing metric structure and curvature response.                                      | D   |
| $P_H$                        | Higgs projection                   | Displacement projection producing Higgs-sector scalar structure.                                             | D   |
| $P_A$                        | Electromagnetic projection         | Compact phase projection producing gauge-like electromagnetic structure.                                     | D   |
| $P_X$                        | Strong-field correction projection | Continuum/self-energy projection producing weak strong-field residuals.                                      | D/C |
| $P_\theta$                   | Cosmological phase projection      | Homogeneous low-energy phase projection associated with expansion response.                                  | D/C |
| $Q_{\mu\nu}$                 | Metric seed tensor                 | Bilinear projection correlation used to construct the effective spacetime metric.                            | D   |
| $g_{\mu\nu}^{\text{eff}}$    | Effective metric                   | Determinant-normalized observable metric recovered from $Q_{\mu\nu}$ .                                       | D   |
| $g_{\mu\nu}^{\text{PR}}$     | PR metric                          | Projection Relativity metric; equals Kerr outside $r_c$ in rotating exterior recovery.                       | D/C |
| $g_{\mu\nu}^{\text{Kerr}}$   | Kerr metric                        | Required exterior rotating recovery target.                                                                  | C   |
| $G_P$                        | Projection Newton coupling         | Low-energy gravitational coupling; constrained by $G_P = G_N$ .                                              | C   |
| $R_{\max}$                   | Maximum curvature scale            | Curvature saturation scale linked to spectral cutoff.                                                        | D/C |
| $r_c$                        | Finite-core radius                 | Radius of saturated projection core replacing divergent singularity.                                         | D/C |
| $\rho_P(\mu^2)$              | Spectral density                   | Weighted internal excitation density controlling propagator self-energy.                                     | D   |
| $F(k^2)$                     | Projection self-energy             | Spectral correction kernel in the propagator.                                                                | D/C |
| $D_{\mu\nu\rho\sigma}^{(P)}$ | Projection propagator              | Effective spin-2 propagator corrected by $F(k^2)$ .                                                          | D/C |
| $\Sigma_X$                   | Strong-field correction            | Weak projection self-energy correction to Kerr/Teukolsky response.                                           | D/C |

| Symbol                | Name                              | Meaning in PR                                                                              | Tag |
|-----------------------|-----------------------------------|--------------------------------------------------------------------------------------------|-----|
| $A_X/A_0$             | Sideband amplitude ratio          | Relative strength of projection-sector ringdown residual.                                  | C   |
| $A(x)$                | Projection amplitude              | Amplitude/displacement coordinate in $\Psi = Ae^{i\theta}$ ; associated with Higgs sector. | D   |
| $\theta(x)$           | Compact phase                     | Internal compact phase coordinate; associated with electromagnetic sector.                 | D   |
| $A_\mu$               | Gauge potential                   | Electromagnetic phase projection.                                                          | D   |
| $D_\mu$               | Gauge-covariant derivative        | Covariant derivative $\partial_\mu - iqA_\mu$ generated by compact phase structure.        | D   |
| $F_{\mu\nu}$          | EM field strength                 | Gauge field tensor $\partial_\mu A_\nu - \partial_\nu A_\mu$ .                             | D   |
| $J_\theta^\mu$        | Phase current                     | Conserved Noether current of compact phase sector.                                         | D   |
| $Z_A$                 | Compact closure factor            | Geometric normalization yielding $\alpha_{\text{PR}}^{-1} = 4\pi Z_A$ .                    | D/C |
| $\alpha_{\text{PR}}$  | Geometric fine-structure constant | Emergent compact phase coupling scale.                                                     | D/C |
| $Z_H$                 | Higgs normalization               | Spectral overlap normalization for Higgs displacement sector.                              | D/C |
| $H(x)$                | Higgs projection observable       | Scalar displacement projection $P_H[\Psi]$ .                                               | D   |
| $H_{\text{PR}}$       | PR Hamiltonian density            | Unified projected energy density.                                                          | D   |
| $S_{\text{PR}}$       | Master action                     | Variational action over spacetime and internal coordinates.                                | D   |
| $\theta_{\text{hom}}$ | Homogeneous phase mode            | Large-scale low-energy mode associated with cosmological expansion.                        | D/C |
| $H_{\text{eff}}$      | Effective Hubble rate             | Expansion response generated by homogeneous phase drift.                                   | D/C |
| $B_{\text{PR}}$       | Residual phase coherence scale    | Predicted nanogauss-scale compact phase coherence observable.                              | C   |

## 2 Master Chain and Projection Correspondence

The universal structure of PR is the projection chain

$$\boxed{\Psi \longrightarrow O_X \longrightarrow \{u_n, \lambda_n\} \longrightarrow P_i[\Psi] \longrightarrow O_i} \quad (2)$$

The full observable-sector correspondence is

$$\boxed{\Psi \rightarrow O_X \rightarrow \{u_n, \lambda_n\} \rightarrow \{P_g, P_H, P_A, P_X, P_\theta\} \rightarrow \{g_{\mu\nu}, H, A_\mu, \Sigma_X, H_{\text{eff}}\}.} \quad (3)$$

Equivalently,

$$\boxed{O = \mathcal{P}[\Psi, O_X, \{u_n, \lambda_n\}].} \quad (4)$$

The native interpretation is:

gravity = stiffness projection,      Higgs = displacement projection,      electromagnetism = compact phase projection

### 3 Internal Hilbert Space

The master field belongs to an internal projection Hilbert space:

$$\Psi \in \mathcal{H}_P. \quad (5)$$

The internal inner product is

$$\langle f|g \rangle = \int d\xi f^*(\xi)g(\xi). \quad (6)$$

For the reduced one-dimensional internal coordinate:

$$\mathcal{H}_w = L^2(\mathbb{R}, dw), \quad \langle f|g \rangle = \int_{\mathbb{R}} dw f^*(w)g(w). \quad (7)$$

The spectral basis satisfies

$$O_X|u_n\rangle = \lambda_n|u_n\rangle, \quad (8)$$

$$\langle u_m|u_n\rangle = \delta_{mn}, \quad (9)$$

and

$$\boxed{\sum_n |u_n\rangle\langle u_n| = \mathbf{1}.} \quad (10)$$

The projection field expands as

$$\boxed{\Psi(x, \xi) = \sum_n c_n(x)u_n(\xi).} \quad (11)$$

### 4 Master Spectral Operator

The internal spectral operator is

$$\boxed{O_X = -\frac{d^2}{dw^2} + V_X(w).} \quad (12)$$

The validation-layer potential is

$$V_X(w) = \Lambda_X^2 \left(1 + a_2 w^2 + a_4 w^4\right), \quad a_2 > 0, \quad a_4 > 0. \quad (13)$$

Its curvature is

$$\frac{d^2 V_X}{dw^2} = \Lambda_X^2 \left(2a_2 + 12a_4 w^2\right) > 0. \quad (14)$$

The eigenvalue equation is

$$O_X u_n(w) = \lambda_n u_n(w). \quad (15)$$

The spectral gaps are

$$\Delta\lambda_n = \lambda_{n+1} - \lambda_n. \quad (16)$$

The first spectral gap is

$$\boxed{\mu_{\min}^2 = \lambda_1 - \lambda_0.} \quad (17)$$

The reference validation values are

$$\lambda_0 \simeq 2.322863529580, \quad \lambda_1 \simeq 5.375830272676, \quad (18)$$

giving

$$\mu_{\min}^2 \simeq 3.052966743096. \quad (19)$$

The spectral gap propagates into the recovery hierarchy:

$$\boxed{\mu_{\min}^2 \rightarrow R_{\max} \rightarrow r_c \rightarrow F(k^2)}. \quad (20)$$

## 5 Master Action

The full PR framework is generated by the spectral–projection action

$$\boxed{S_{\text{PR}} = \int d^4x d\xi \sqrt{-g_{\text{eff}}} \left[ \frac{1}{2} G^{AB} D_A \Psi^\dagger D_B \Psi - V_{\text{PR}}(\Psi, \xi) + \mathcal{L}_{\text{src}} \right]}. \quad (21)$$

Here:

$$\begin{aligned} G^{AB} &= \text{internal manifold metric}, & D_A &= \text{projection covariant derivative}, \\ V_{\text{PR}} &= \text{internal projection potential}, & \mathcal{L}_{\text{src}} &= \text{observable source sector}. \end{aligned}$$

The Euler–Lagrange equation schematically takes the form

$$\frac{1}{\sqrt{-g_{\text{eff}}}} D_A \left( \sqrt{-g_{\text{eff}}} G^{AB} D_B \Psi \right) + \frac{\partial V_{\text{PR}}}{\partial \Psi^\dagger} = \frac{\partial \mathcal{L}_{\text{src}}}{\partial \Psi^\dagger}. \quad (22)$$

Thus the spectral structure, projection dynamics, source coupling, propagator response, and emergent metric are all treated as consequences of the same action principle.

## 6 Emergent Metric and Weak Gravity Recovery

The metric seed tensor is

$$Q_{\mu\nu} = \langle \partial_\mu \Psi \partial_\nu \Psi \rangle. \quad (23)$$

A more explicit internal metric form is

$$Q_{\mu\nu} = \left\langle \partial_\mu \Psi^A \partial_\nu \Psi^B \right\rangle G_{AB}. \quad (24)$$

The observable metric is determinant normalized:

$$\boxed{g_{\mu\nu}^{\text{eff}} = \frac{Q_{\mu\nu}}{|\det Q|^{1/4}}}. \quad (25)$$

The weak-field expansion is

$$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1. \quad (26)$$

The low-energy gravitational recovery condition is

$$G_{\mu\nu}^{\text{PR}} \rightarrow 8\pi G_P T_{\mu\nu}, \quad (27)$$

with

$$\boxed{G_P = G_N.} \quad (28)$$

The massless graviton and weak-field recovery constraints are

$$\boxed{F(0) = 0, \quad \left| \frac{F(k^2)}{k^2} \right| \ll 1 \quad (k^2 \ll \mu_{\min}^2).} \quad (29)$$

## 7 Finite-Core Regularization

Classical singular behavior is replaced by bounded spectral saturation. The finite-core radius is

$$\boxed{r_c = \left( \frac{G_P M}{c^2 R_{\max}} \right)^{1/3}.} \quad (30)$$

The core hierarchy is

$$\mu_{\min}^2 \rightarrow R_{\max} \rightarrow r_c. \quad (31)$$

The interpretation is:

$$r < r_c : \text{ saturated projection core,} \quad r > r_c : \text{ GR/Kerr exterior recovery.}$$

The curvature is bounded schematically by

$$|R| \lesssim R_{\max}. \quad (32)$$

## 8 Kerr Exterior Recovery

The rotating exterior recovery condition is

$$\boxed{g_{\mu\nu}^{\text{PR}} = g_{\mu\nu}^{\text{Kerr}}, \quad r > r_c.} \quad (33)$$

The Kerr structure functions are

$$\Sigma_{\text{Kerr}} = r^2 + a^2 \cos^2 \theta, \quad (34)$$

$$\Delta_{\text{Kerr}} = r^2 - \frac{2G_P M}{c^2} r + a^2. \quad (35)$$

The PR structure functions are

$$\Sigma_{\text{PR}} = \Sigma_{\text{Kerr}} + \delta\Sigma(r), \quad (36)$$

$$\Delta_{\text{PR}} = \Delta_{\text{Kerr}} + \delta(r). \quad (37)$$

Exterior recovery requires

$$\delta\Sigma(r) \rightarrow 0, \quad \delta(r) \rightarrow 0, \quad r \gg r_c. \quad (38)$$

## 9 Projection Propagator and Self-Energy

The spectral density is

$$\rho_P(\mu^2) = \sum_n \left| \langle u_n | \hat{V} | u_0 \rangle \right|^2 \delta(\mu^2 - \mu_n^2). \quad (39)$$

Normalization:

$$\int d\mu^2 \rho_P(\mu^2) = 1. \quad (40)$$

The self-energy kernel is

$$F(k^2) = \int d\mu^2 \frac{\rho_P(\mu^2)}{k^2 - \mu^2 + i\epsilon}. \quad (41)$$

The projection propagator is

$$D_{\mu\nu\rho\sigma}^{(P)}(k) = \frac{P_{\mu\nu\rho\sigma}^{(2)}}{k^2 - F(k^2) + i\epsilon}. \quad (42)$$

The stability requirement is

$$k^2 - F(k^2) > 0 \quad (43)$$

in the physical low-energy recovery regime.

## 10 Gravitational Waves and Ringdown

Projection gravitational waves are perturbations of the emergent metric:

$$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + h_{\mu\nu}. \quad (44)$$

The transverse-traceless propagation equation is

$$[\square + \Pi_X(\square)] h_{\mu\nu}^{\text{TT}} = 16\pi G_P T_{\mu\nu}^{\text{TT}}. \quad (45)$$

The Kerr–Teukolsky inheritance structure is

$$O_{\text{PR}} = O_{\text{Teuk}} + \Sigma_X. \quad (46)$$

The weak correction condition is

$$|\Sigma_X| \ll |O_{\text{Teuk}}|. \quad (47)$$

The ringdown response may be decomposed as

$$A(\omega) = A_{\text{Kerr}}(\omega) + A_X(\omega), \quad (48)$$

with

$$A_X(\omega) = \int d\mu \frac{\rho_X(\mu)}{\omega - \mu + i\Gamma_X(\mu)}. \quad (49)$$

The projected sideband amplitude scale is

$$A_X/A_0 \sim 10^{-3}. \quad (50)$$

## 11 Higgs Displacement Sector

The amplitude–phase decomposition is

$$\boxed{\Psi(x) = A(x)e^{i\theta(x)}.} \quad (51)$$

Differentiation gives

$$\partial_\mu \Psi = e^{i\theta} (\partial_\mu A + iA\partial_\mu \theta). \quad (52)$$

The kinetic norm separates:

$$\boxed{|\partial_\mu \Psi|^2 = (\partial_\mu A)(\partial^\mu A) + A^2(\partial_\mu \theta)(\partial^\mu \theta).} \quad (53)$$

The Higgs projection is

$$\boxed{H(x) = P_H[\Psi] = A(x).} \quad (54)$$

The Higgs-sector Lagrangian is

$$\mathcal{L}_H = \frac{1}{2}(\partial_\mu A)(\partial^\mu A) - V(A). \quad (55)$$

The potential is

$$V(A) = \alpha_H A^2 + \beta_H A^4, \quad \beta_H > 0. \quad (56)$$

The vacuum condition is

$$\frac{dV}{dA} = 2\alpha_H A + 4\beta_H A^3 = 0, \quad (57)$$

giving

$$A_0^2 = -\frac{\alpha_H}{2\beta_H}. \quad (58)$$

The Higgs spectral normalization is

$$\boxed{Z_H = \int d\xi u_0(\xi) P_H[\Psi].} \quad (59)$$

The Higgs mass curvature relation is

$$\boxed{m_H^2 = \left. \frac{d^2 V}{dA^2} \right|_{A=A_0} = 2\alpha_H + 12\beta_H A_0^2.} \quad (60)$$

## 12 Electromagnetic Compact Phase Sector

The electromagnetic projection is

$$\boxed{A_\mu = P_A[\Psi] \sim \partial_\mu \theta.} \quad (61)$$

The compact phase identification is

$$\boxed{\theta \sim \theta + 2\pi n, \quad n \in \mathbb{Z}.} \quad (62)$$

The gauge-covariant derivative is

$$\boxed{D_\mu = \partial_\mu - iqA_\mu.} \quad (63)$$

The compact gauge transformation is

$$\boxed{A_\mu \rightarrow A_\mu + q^{-1} \partial_\mu \alpha.} \quad (64)$$

The electromagnetic field tensor is

$$\boxed{F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.} \quad (65)$$

The Maxwell-type Lagrangian is

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + J_\mu A^\mu. \quad (66)$$

The Maxwell recovery equation is

$$\boxed{\partial_\mu F^{\mu\nu} = J^\nu.} \quad (67)$$

The compact phase Lagrangian is

$$\mathcal{L}_A = \frac{1}{2} A^2 (\partial_\mu \theta) (\partial^\mu \theta). \quad (68)$$

The Noether current is

$$\boxed{J_\theta^\mu = A^2 \partial^\mu \theta.} \quad (69)$$

The conservation law is

$$\boxed{\partial_\mu J_\theta^\mu = 0.} \quad (70)$$

## 13 Geometric Fine-Structure Closure

The fine-structure closure chain is

$$\boxed{O_X \rightarrow \{u_n, \lambda_n\} \rightarrow \mu_{\min} \rightarrow P_A[\Psi] \rightarrow Z_A \rightarrow \alpha_{\text{PR}}^{-1}.} \quad (71)$$

The compact closure factor is

$$Z_A = \oint d\theta u_0(\theta) P_A[\Psi]. \quad (72)$$

The geometric closure relation is

$$\boxed{\alpha_{\text{PR}}^{-1} = 4\pi Z_A.} \quad (73)$$

The validated effective value is

$$\boxed{\alpha_{\text{PR}}^{-1} \simeq 137.0365.} \quad (74)$$

The stability chain is

$$\delta\alpha \rightarrow \delta F_{\text{eff}}(k^2) \rightarrow \delta\Sigma_X \rightarrow \text{loss of spectral regularity}. \quad (75)$$

## 14 Unified Projection Energy

The projection-sector Lagrangian is

$$\mathcal{L}_{\text{PR}} = \frac{1}{2} (\partial_\mu A) (\partial^\mu A) + \frac{1}{2} A^2 (\partial_\mu \theta) (\partial^\mu \theta) - V(A) - \frac{1}{2} \alpha_g P_g[\Psi]^2. \quad (76)$$

Canonical momenta:

$$\Pi_A = \frac{\partial \mathcal{L}_{\text{PR}}}{\partial \dot{A}} = \dot{A}, \quad (77)$$

$$\Pi_\theta = \frac{\partial \mathcal{L}_{\text{PR}}}{\partial \dot{\theta}} = A^2 \dot{\theta}. \quad (78)$$

The Hamiltonian density is

$$H_{\text{PR}} = \Pi_A \dot{A} + \Pi_\theta \dot{\theta} - \mathcal{L}_{\text{PR}}. \quad (79)$$

Explicitly:

$$H_{\text{PR}} = \frac{1}{2} \dot{A}^2 + \frac{1}{2} (\nabla A)^2 + \frac{1}{2} A^2 \dot{\theta}^2 + \frac{1}{2} A^2 (\nabla \theta)^2 + V(A) + \frac{1}{2} \alpha_g P_g[\Psi]^2. \quad (80)$$

The sector decomposition is

$$H_{\text{PR}} = H_g + H_H + H + H_{\text{vac}}. \quad (81)$$

The observable relativistic energy relation is

$$Mc^2 = E_{\text{PR}} = \int d^3x H_{\text{PR}}. \quad (82)$$

## 15 Projection Unitarity and Information Preservation

The internal spectral evolution equation is

$$i \frac{\partial \Psi}{\partial t} = \hat{H}_P \Psi. \quad (83)$$

The projection Hamiltonian is

$$\hat{H}_P = O_X + \hat{V}_f. \quad (84)$$

The unitary evolution operator is

$$U(t) = e^{-i\hat{H}_P t}. \quad (85)$$

Unitarity:

$$U^\dagger U = I. \quad (86)$$

Norm conservation:

$$\langle \Psi | \Psi \rangle = 1. \quad (87)$$

Collapse is interpreted as

$$\text{collapse} \rightarrow \text{spectral excitation redistribution} \rightarrow \text{finite-core saturation}. \quad (88)$$

## 16 Cosmological Projection Structure

The cosmological chain is

$$\boxed{\Psi \rightarrow \theta_{\text{hom}} \rightarrow J_{\text{vac}} \rightarrow H_{\text{eff}}(t)}. \quad (89)$$

The homogeneous phase Lagrangian is

$$\mathcal{L}_{\text{hom}} = \frac{1}{2} f_{\theta}^2 \dot{\theta}_{\text{hom}}^2 - V(\theta_{\text{hom}}). \quad (90)$$

The homogeneous phase potential is

$$V(\theta_{\text{hom}}) = \Lambda_{\theta}^4 \left[ 1 - \cos \left( \frac{\theta_{\text{hom}}}{f_{\theta}} \right) \right]. \quad (91)$$

Small-displacement expansion:

$$V(\theta_{\text{hom}}) \simeq \frac{1}{2} m_{\theta}^2 \theta_{\text{hom}}^2, \quad m_{\theta}^2 = \frac{\Lambda_{\theta}^4}{f_{\theta}^2}. \quad (92)$$

The vacuum source rule is

$$J_{\text{vac}} \propto \langle T^{00} \rangle_{\text{hom}}. \quad (93)$$

The source selection hierarchy is

$$\boxed{\langle T^{00} \rangle_{\text{hom}} \rightarrow \theta_{\text{hom}}, \quad T^{00} - \langle T^{00} \rangle_{\text{hom}} \rightarrow P_g}. \quad (94)$$

The emergent Hubble rate is

$$\boxed{H_{\text{eff}} = A_{\theta} \dot{\theta}_{\text{hom}}}. \quad (95)$$

The cosmological response equation is

$$\boxed{\ddot{\theta}_{\text{hom}} + 3H_{\text{eff}} \dot{\theta}_{\text{hom}} + \frac{dV}{d\theta_{\text{hom}}} = J_{\text{vac}}}. \quad (96)$$

The late-time recovery structure is

$$H_{\text{eff}}(z) = H_{\text{GR}}(z) + \delta H_{\text{PR}}(z), \quad \left| \frac{\delta H_{\text{PR}}}{H_{\text{GR}}} \right| \ll 1. \quad (97)$$

## 17 Observational Structure

The observational hierarchy is

$$\boxed{\mu_{\text{min}} \rightarrow F(k^2) \rightarrow \Sigma_X \rightarrow O_{\text{obs}}}. \quad (98)$$

Weak-field recovery:

$$g_{\mu\nu}^{\text{PR}} = g_{\mu\nu}^{\text{GR}}, \quad r \gg r_c. \quad (99)$$

Kerr recovery:

$$g_{\mu\nu}^{\text{PR}} = g_{\mu\nu}^{\text{Kerr}}, \quad r > r_c. \quad (100)$$

Strong-field condition:

$$k^2 \sim \mu_{\min}^2. \quad (101)$$

Projection correction amplitude:

$$\boxed{A_X/A_0 \sim 10^{-3}.} \quad (102)$$

Residual phase-coherence scale:

$$\boxed{B_{\text{PR}} \sim 10^{-12} \text{ T} \sim 10 \text{ nG}.} \quad (103)$$

## 18 Validation and Completion Status

| Framework Sector             | Status                        | Validation Logic                                                             |
|------------------------------|-------------------------------|------------------------------------------------------------------------------|
| Internal spectral operator   | Closed at framework level     | Eigenvalue problem and spectral gap define internal scale.                   |
| Emergent metric              | Closed at framework level     | Determinant-normalized metric seed recovers weak-field geometry.             |
| Weak-field GR recovery       | Constrained and validated     | Requires $G_P = G_N$ , $F(0) = 0$ , and $ F/k^2  \ll 1$ .                    |
| Kerr exterior recovery       | Constrained and validated     | Requires $g_{\mu\nu}^{\text{PR}} = g_{\mu\nu}^{\text{Kerr}}$ for $r > r_c$ . |
| Finite-core structure        | Derived/constrained           | Spectral gap sets $R_{\max}$ and finite-core radius $r_c$ .                  |
| Projection propagator        | Closed at effective level     | Spectral self-energy remains suppressed at low energy.                       |
| Higgs displacement sector    | First-order closed            | Amplitude projection generates displacement sector and mass curvature.       |
| Electromagnetic phase sector | First-order closed            | Compact phase projection generates gauge-like sector and Noether current.    |
| Fine-structure closure       | First-order geometric closure | $\alpha_{\text{PR}}^{-1} = 4\pi Z_A$ gives validated closure value.          |
| Unitarity/information        | Effective-level closed        | Hilbert evolution preserves norm and redistributes information spectrally.   |
| Cosmological projection      | Effective regime              | Homogeneous phase drift provides late-time projection response.              |
| Strong-field sidebands       | Predictive                    | Weak continuum correction scale $A_X/A_0 \sim 10^{-3}$ .                     |
| Full matter-sector extension | Future development            | Additional projection sectors may extend the framework.                      |

| Framework Sector              | Status             | Validation Logic                                                                            |
|-------------------------------|--------------------|---------------------------------------------------------------------------------------------|
| Nonperturbative UV completion | Future development | Full high-energy completion remains an extension beyond the current effective architecture. |

## 19 Single-Line Native Definitions

| Concept                 | PR Native Definition                                                           |
|-------------------------|--------------------------------------------------------------------------------|
| Spacetime               | The low-energy metric projection of the internal spectral manifold.            |
| Gravity                 | The stiffness projection of the internal manifold.                             |
| Higgs structure         | The displacement/amplitude projection of the manifold.                         |
| Electromagnetism        | The compact phase projection of the manifold.                                  |
| Fine-structure constant | The compact geometric closure scale of the electromagnetic phase sector.       |
| Black-hole interior     | A finite saturated projection-core region rather than a divergent singularity. |
| Kerr exterior           | The recovered rotating low-energy exterior limit outside $r_c$ .               |
| Ringdown correction     | A weak internal continuum response added to the Kerr–Teukolsky structure.      |
| Cosmological expansion  | Homogeneous phase drift of the projection manifold.                            |
| Energy                  | The low-energy exterior projection of conserved internal spectral energy.      |